

DV033G, Software Testing

Assignment 1, analysis

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Part A – Fundamentals (M1)

Activity 1.1 – Quality-attribute analysis (ISO/IEC 25010)

We choose following three features:

- FR-1.1 Registration
- FR-3.2 State machine
- FR-5.2 Robustness and fallback

Feature	ISO/IEC 25010	Justification	Quality requirement
FR-1.1 Registration	3.4.4 User error protection	We want to prevent incorrect user input with clear validation / format rules.	Given incorrect email (e.g. example.com) the system must not proceed and must show an error message to a user, clearly explaining the issue.
	3.4.3 Operability	We want to provide clear form guidance (including labels and placeholder hints) and feedback when input is incorrect.	
FR-3.2 State machine	3.1.2 Functional correctness	Only legal transitions between states must be enforced, while illegal transitions must be rejected.	Given a listing status CANCELLED , a request to transition it to IN_PROGRESS must be rejected and the listing's status must remain CANCELLED .
	3.6.2 Integrity	The system should prevent illegal transitions of the listing's state, e.g. from unauthorized API calls.	

Feature	ISO/IEC 25010	Justification	Quality requirement
FR-5.2 Robustness and fallback	3.5.3 Fault tolerance	The system should fall back to a standard, deterministic answer instead of throwing an exception.	Given a listing profile with non-empty description. The prompt builder must encapsulate the description in untrusted-data delimiters.
	3.6.6 Resistance	The risk of prompt-injection should be minimized by using appropriate techniques, e.g. untrusted-data delimiters.	

Activity 1.2 – Bug analysis (error → fault → failure)

Error	The developer made a mistake in one of the checks that prevents a seeker with NEW status to have more than one active booking.
Fault	The logic that validates maximum allowed bookings for NEW seekers is broken.
Failure	A seeker with a NEW status may have few active bookings at the same time.

The code responsible for the failure is found in `BookingService::createBooking`:

```
// Rule 2: seeker's active bookings below the trust-tier max (FR-4.4 rule 2).
long seekerActive = activeBookingCountForSeeker(seekerId);
if (seekerActive > seeker.getMaxConcurrentBookings()) {
    throw new BookingRejectedException("Seeker booking limit reached for tier " +
    seeker.getTrustTier());
}
```

The problem here is exclusive comparison, instead of `>` the developer should have put `>=`. This problem should have been caught on the unit test level, because it is a part of a simple, fundamental part of the logic on the code level (a simple `if` statement in this case).

Part B – Specification-based design (M2)

Activity 2.1 – Equivalence Partitioning

Email

Partition	Meaning	Representative
@		
Invalid low	There is no @ sign in an email.	example.com
Valid	There is only one @ sign in an email.	example@domain.com
Invalid high	There is more than one @ sign in an email.	example@example@example.domain.com
Local part		
Invalid	There is no local part.	@domain.com
Valid	There is a non-empty local part.	example@domain.com
.		
Invalid	There is no .	example@domain
Valid	There is one or more .	example@domain.com
Length		

Invalid low	Email length is less than 5 characters	a@b.
Valid	Email length between 5 and 254 inclusive.	example@domain.com
Invalid high	Email length above 254 characters.	abcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyz012345678912@abcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyz0123456789.abcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyz0123456789.abcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyz0123456789.abcdefghijklmnopqrstuvwxyz0123456789.abcdefghijklmnopqrstuvwxyz0123456789.com

Display name

Partition	Meaning	Representative
Display name length		
Invalid low	Display name length less than 2 characters.	a
Valid	Display name length between 2 and 40 characters inclusive.	Sergio
Invalid high	Display name length above 40 characters.	Alexander_The_Great_conqueror_Of_Worlds89
Special characters		

Valid	The display name contains only letters/spaces/hyphens/apostrophes.	Sergio
Invalid	The display name contains a non-alphanumeric character.	Sergio <3

Phone number

Partition	Meaning	Representative
07 format		
Invalid low	The phone number (including 07) is less than 10 digits long.	070
Valid	The phone number (including 07) is 10 digits long.	0712345678
Invalid high	The phone number length is above 10 digits.	07123456789
+46 format		
Invalid low	The phone number (including +46) is less than 12 characters long.	+467
Valid	The phone number (including +46) is 12 characters long.	+46723456789
Invalid high	The phone number length is above 12 characters.	+467234567890

07 format (special characters)		
Invalid	The phone number contains non-digit characters.	07abc
Valid	The phone number only contains digits, starting with 07.	0712345678
+46 format (special characters)		
Invalid	The phone numbers contain one + character and any other non-digit(s).	+467abc
Valid	The phone number contains exactly one + character followed by 467 and digits.	+46723456789

Wallet (top-up)

Partition	Meaning	Representative
Top-up		
Invalid low	Top-up below 10.00 SEK	9.99 SEK
Valid	Top-up between 10.00 and 5000.00 SEK	11.00 SEK
Invalid high	Top-up above 5000.00 SEK	5000.01 SEK
Resulting balance		

Invalid low	The balance is below 0.00 SEK	-21000.00 SEK
Valid	The balance is between 0.00 and 20000.00 SEK	10.00 SEK
Invalid high	The balance is above 20000.00 SEK	4000.00 SEK (when the current balance is 17000.00 SEK)

Selected representatives per class

Class	Input	Expected outcome
@	example@example@example.domain.com	Rejected
Local part	example@domain.com	Accepted
.	example@domain.com	Accepted
Email length	abcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyz012345678912@abcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyz0123456789.abcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyz0123456789.abcdefghijklmnopqrstuvwxyzabcdefghijklmnopqrstuvwxyz0123456789.com	Rejected
Display name length	Alexander_The_Great_conqueror_Of_Worlds89	Rejected
Special characters	Sergio <3	Rejected
07 format (length)	0712345678	Accepted
+46 format (length)	+46723456789	Accepted

07 format (special characters)	0712345678	Accepted
+46 format (special characters)	+46723456789	Accepted
Top-up	5000.01 SEK	Rejected
Resulting balance	4000.00 SEK	Rejected

Activity 2.2 – Boundary Value Analysis

Purpose	Value	Expected result
Just below lower top-up boundary	9.99 SEK	Reject
Lower top-up boundary	10.00 SEK	Accept
Just above lower top-up boundary	10.01 SEK	Accept
Just below upper top-up boundary	4999.99 SEK	Accept
Upper top-up boundary	5000.00 SEK	Accept
Just above upper top-up boundary	5000.01 SEK	Reject
Just below maximum balance	5000.00 + 5000.00 + 5000.00 + 4999.99 SEK	Accept
Maximum balance	5000.00 + 5000.00 + 5000.00 +	Accept

	5000.00 SEK	
Just above maximum balance	$5000.00 + 5000.00 + 5000.00 + 4999.99 + 0000.02$ SEK	Reject

Activity 2.3 – Decision table (trust tier → limits)

Trust tier	Max concurrent active bookings	Platform fee
NEW	1	15%
VERIFIED	3	12%
TRUSTED	5	8%
PRO_SITTER	10	5%